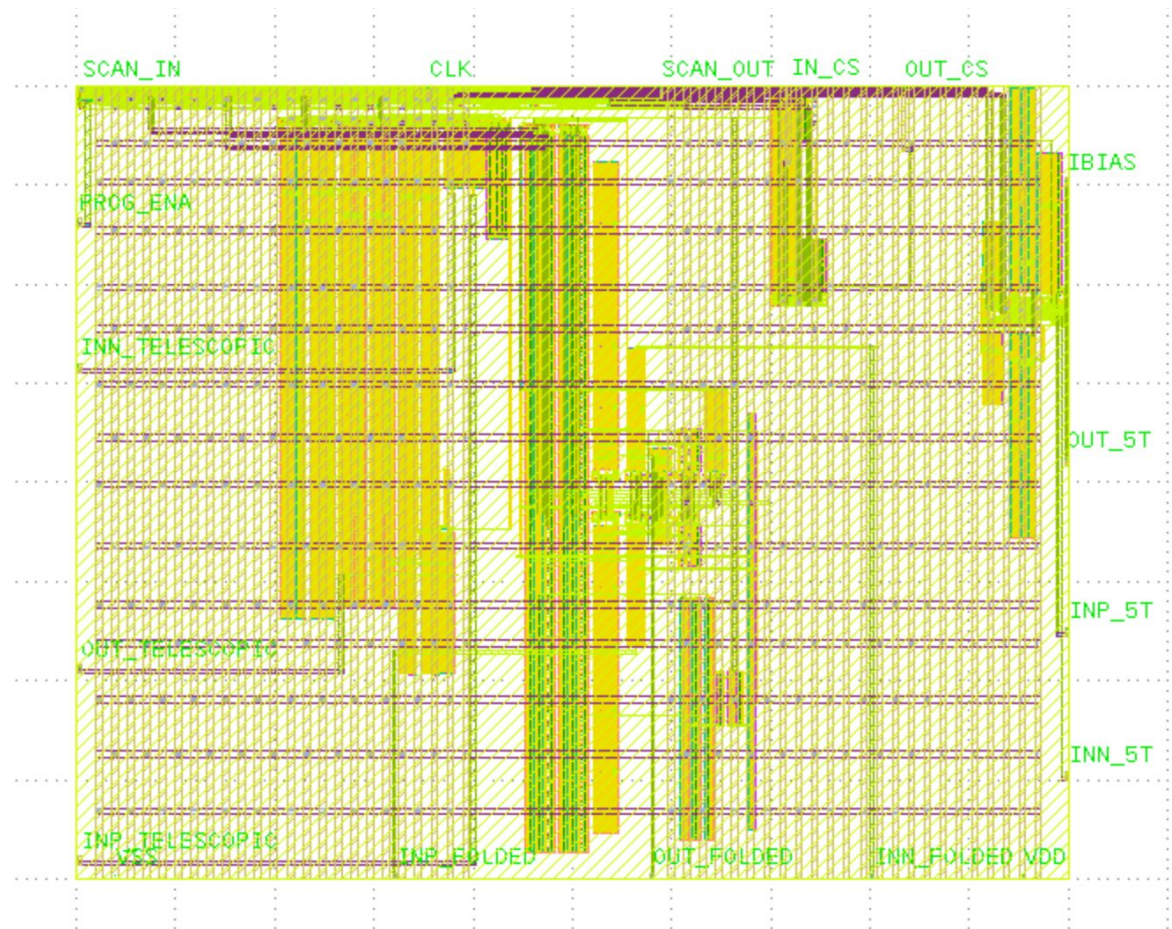


C1 Four Topology MOSbius

Final Layout 08/16

Discord name	Affiliation	Experience	Role
elij0100	Columbia University	Undergraduate	Team Lead
boibun.	Columbia University	Undergraduate	Team Member
maxdrucker_83950	Columbia University	Undergraduate	Team Member
Manwell	Columbia University	Undergraduate	Team Member

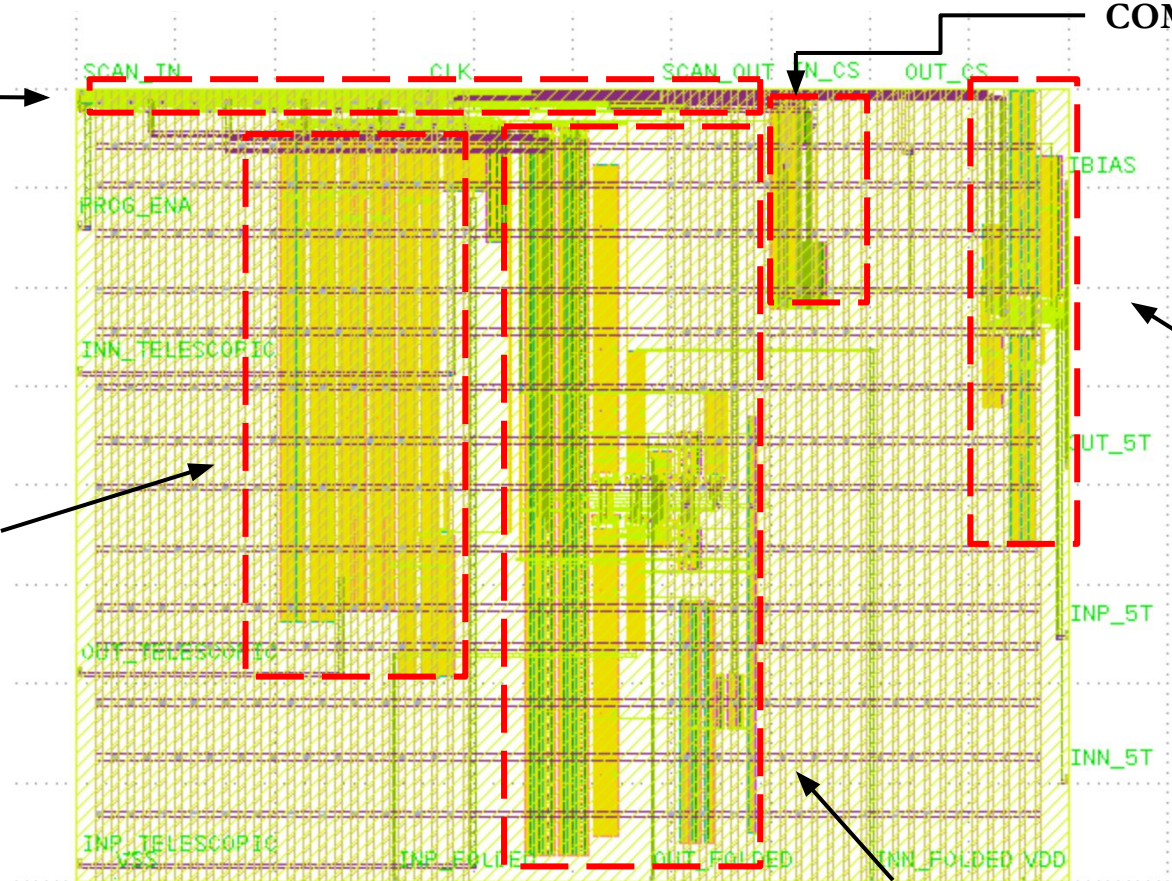


SCAN
CHAIN

COMMON-SOURCE
& BIAS

TELESCOPIC

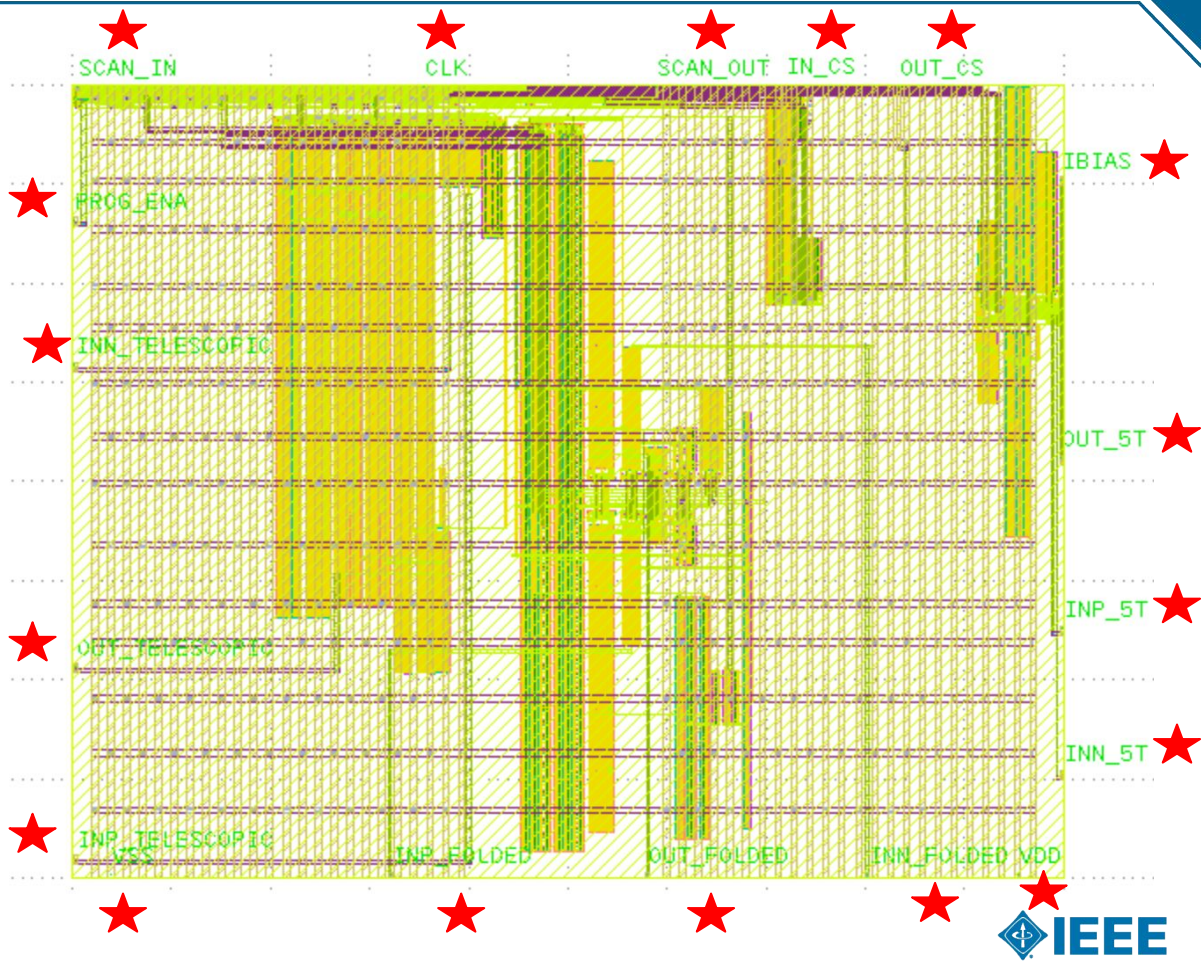
5-TRANSISTOR

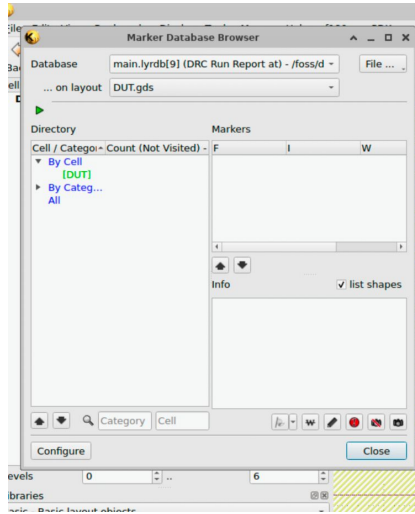


FOLDED CASCODE

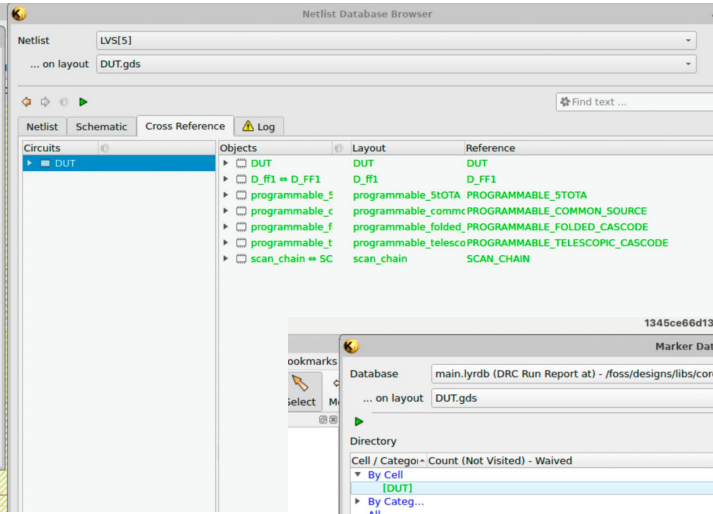
- $500\mu\text{m} \times 400\mu\text{m}$
- 18 total pins: 5 on top, 5 on bottom, 4 on left, 4 on right
- Pins are distributed along the edges to simplify wire bonding
- All routing to pins uses wide top-level metal traces

★ = PIN

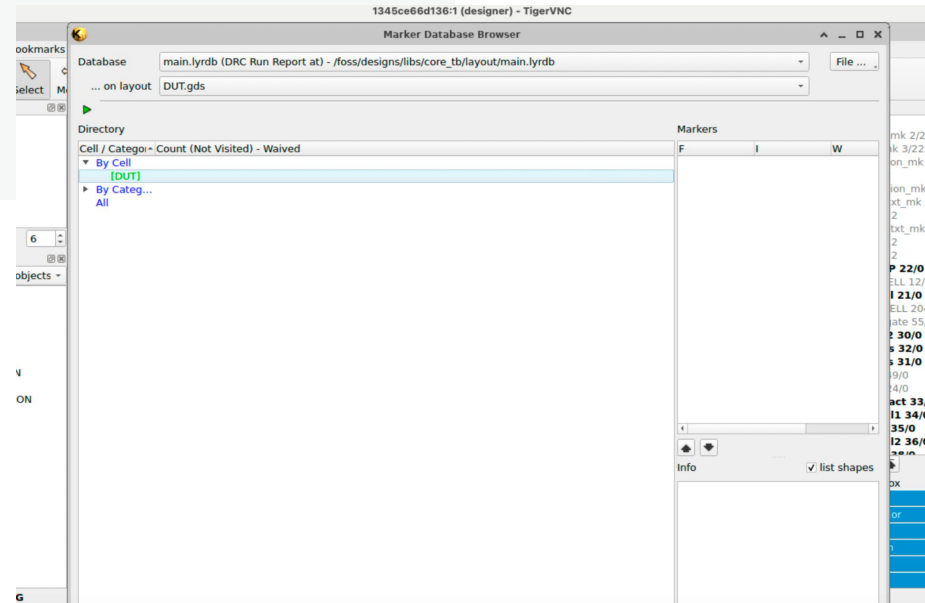




DRC

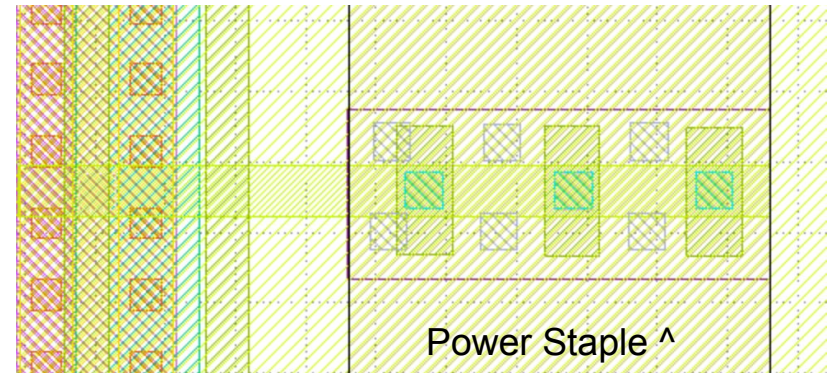
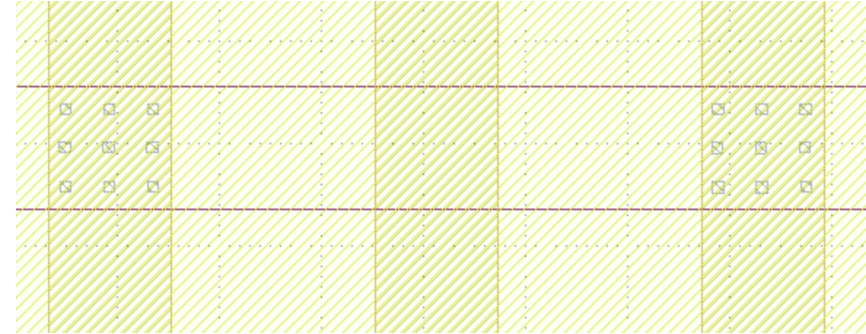


LVS

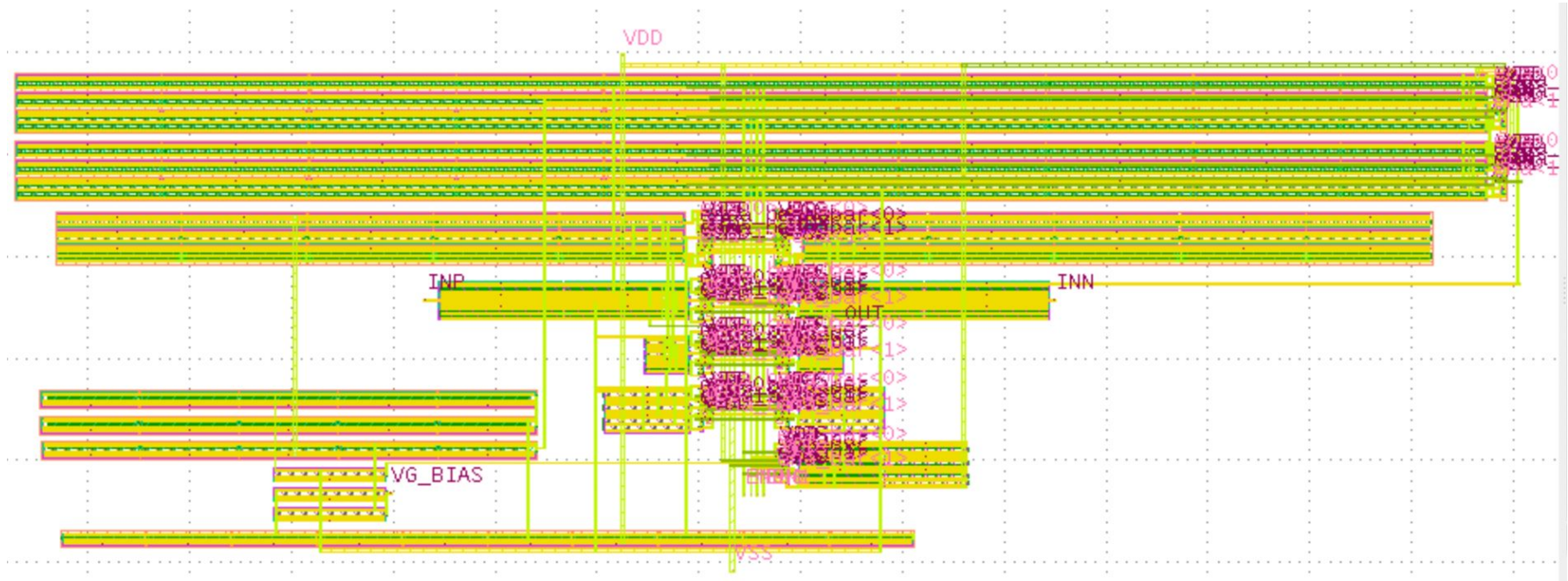


Power Grid

- Power grid uses $3\mu\text{m}$ thick wires on Metal4 and Metal5
- Overlapping Metal4/Metal5 regions fit 9 vias each, significantly reducing parasitic resistance and ensuring power integrity
- "Power staples" use the same approach to deliver power to blocks within cells
- Grid distributes power evenly across the entire chip
- Overlapping metal layers provide metal-to-metal decoupling capacitance
- Wide wires limit current density to avoid electromigration



Folded Cascode Layout



Folded Cascode Layout Considerations

- Matched devices placed with mirror symmetry, all gates running the same direction so both devices see the same process conditions
- Longer L on current mirrors and current sources ($L = 1\mu\text{m}$) for better threshold voltage matching
- Left and right branches placed symmetrically, aside from the PMOS current sources
- $1.4\mu\text{m}+$ gap between N-well and P-sub regions for noise isolation between PMOS and NMOS groups (also a design rule requirement)
- Bulk ties on all transistors, giving a low-impedance path from the body to a clean supply (VDD for PMOS, VSS for NMOS) to reduce substrate noise coupling
- Routing convention: Metal1 horizontal, Metal2 vertical, Metal3 horizontal for enable and bias buses

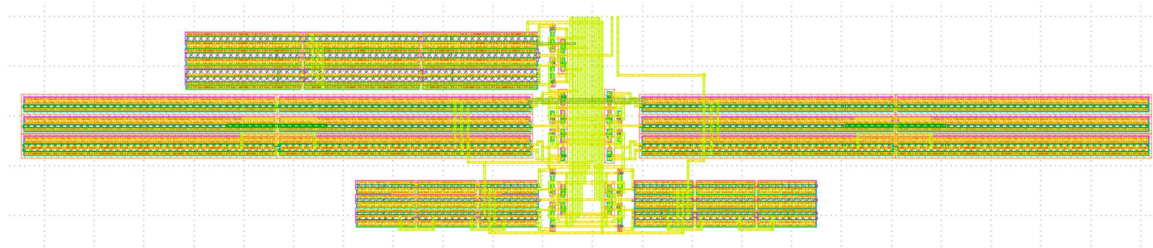
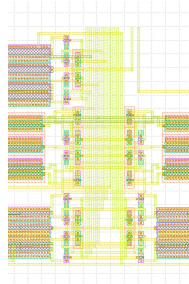
Telescopic Cascode Layout

- Telescopic cascode layout has enable signals routed along the bottom edge
- Matched transistor pairs placed side by side with the same orientation for best matching during fabrication
- Metal2 used for longer routing runs over Metal1 to reduce parasitic resistance
- Bias network connected in the top left corner, grouped with all non-enable analog pins
- Analog signal pins kept physically separated from digital enable pins to minimize noise coupling



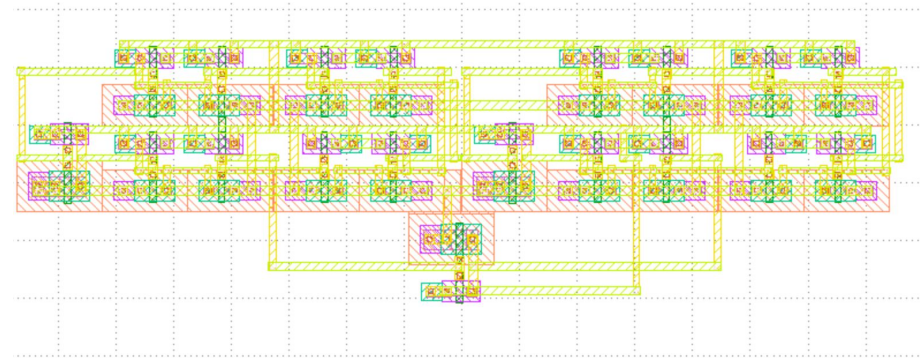
5 Transistor

- Programmable transistors placed face to face to simplify routing and reduce interconnect length
- Enable, input, and bias pins placed along the top edge for clean external connection
- Symmetric placement used to improve matching between devices
- Vertical and centralized routing for an organized layout
- DRC clean, LVS clean
- Cell dimensions: $226\mu\text{m} \times 43\mu\text{m}$



Digital

- Scan chain consists of 9 D flip-flops
- Each DFF is built from two D latches and a clock inverter
- Metal hopping used on gate connections to avoid antenna effects
- DFFs stacked left to right, aligned with the Metal2 routing direction for efficient long-path connections



Thank You!
Any feedback is greatly appreciated.